12TH BIOLOGY · CHAPTER 17 · FULL CHAPTER MCQS

Human Reproductive Systems

48 topic-wise MCQs from the complete chapter with answers and reasons

17.1.1 MALE REPRODUCTIVE SYSTEM

1. The reproductive system becomes functional at:

(a) Birth
(b) Puberty
(c) Menopause
(d) Embryonic stage

Answer: (b) Puberty — Other systems are functional since or shortly after birth.

2. The skin pouch that contains the testes is the:

(a) Epididymis
(b) Prostate
(c) Scrotum
(d) Inguinal canal

Answer: (c) Scrotum — It lies outside the abdomen.

3. The scrotum maintains the testes at about:

(a) 30°C
(b) 40°C
(c) 37°C
(d) 34°C

Answer: (d) 34°C — This is slightly lower than core body temperature.

4. The length of a testis is about:

(a) 3–5 cm
(b) 1–2 cm
(c) 8–10 cm
(d) 12–15 cm

Answer: (a) 3–5 cm — Its width is 2–3 cm.

5. A testis is composed of highly convoluted:

(a) Oviducts
(b) Seminiferous tubules
(c) Ejaculatory ducts
(d) Ovarian follicles

Answer: (b) Seminiferous tubules — Sperms are formed in them.

6. The cells that nourish the developing sperms are:

(a) Sertoli cells
(b) Leydig cells
(c) Germline cells
(d) Spermatids

Answer: (a) Sertoli cells — They lie in the germinal epithelium.

7. From the seminiferous tubules, sperms first enter the:

(a) Epididymis
(b) Urethra
(c) Vas deferens
(d) Rete testis

Answer: (d) Rete testis — It is a network of tubules.

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8. Sperms become motile in the:

(a) Seminiferous tubules (b) Urethra
(c) Epididymis (d) Prostate gland

Answer: (c) Epididymis — They stay there for at least 18 hours.

9. The vas deferens passes into the abdominal cavity via the:

(a) Urethra (b) Inguinal canal
(c) Cervix (d) Rete testis

Answer: (b) Inguinal canal — Then it joins the duct of a seminal vesicle.

10. The vas deferens joins the duct of the seminal vesicle to form the:

(a) Urethra (b) Epididymis
(c) Ejaculatory duct (d) Rete testis

Answer: (c) Ejaculatory duct — It passes through the prostate gland.

11. The prostate gland produces about what part of the fluid of semen?

(a) 60% (b) 30%
(c) 10% (d) 90%

Answer: (a) 60% — It lies just below the urinary bladder.

12. The glands that lubricate the urethra for sperm passage are the:

(a) Seminal vesicles (b) Adrenal glands
(c) Prostate glands (d) Bulbourethral glands

Answer: (d) Bulbourethral glands — They also provide a neutral environment.

13. Primary spermatocytes are:

(a) Haploid (b) Triploid
(c) Diploid (d) Tetraploid

Answer: (c) Diploid — They divide by meiosis-I.

14. Secondary spermatocytes are formed by:

(a) Mitosis (b) Meiosis-I
(c) Meiosis-II (d) Maturation

Answer: (b) Meiosis-I — They are haploid (1n).

15. One spermatogonium produces how many spermatids?

(a) Four (b) Two
(c) One (d) Eight

Answer: (a) Four — Each matures into a sperm.

16. The acrosome is formed mainly of:

(a) Mitochondria (b) Nucleus
(c) Centriole (d) Golgi body

Answer: (d) Golgi body — It contains enzymes for fertilization.

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17. The midpiece of a sperm contains:

(a) Acrosome (b) Mitochondria and a centriole
(c) Genetic material (d) Golgi body

Answer: (b) Mitochondria and a centriole — The tail arises from the centriole.

18. The tail of a sperm is formed of:

(a) Mitochondria (b) Golgi body
(c) Microtubules (d) Chromatin

Answer: (c) Microtubules — It is the axial filament.

19. GnRH is released by the:

(a) Hypothalamus (b) Anterior pituitary
(c) Testes (d) Prostate gland

Answer: (a) Hypothalamus — It stimulates the release of FSH and LH.

20. Inhibin is released by:

(a) Leydig cells (b) Anterior pituitary
(c) Germline cells (d) Sertoli cells

Answer: (d) Sertoli cells — It inhibits further FSH secretion.

21. Testosterone inhibits the release of:

(a) FSH only (b) LH and GnRH
(c) Inhibin (d) Estrogen

Answer: (b) LH and GnRH — This is negative feedback.

17.1.2 FEMALE REPRODUCTIVE SYSTEM

22. The functional units of the ovary are the:

(a) Seminiferous tubules (b) Ovarian follicles
(c) Oviducts (d) Corpus luteum

Answer: (b) Ovarian follicles — They nurture immature eggs.

23. The release of an egg cell from a follicle is called:

(a) Menstruation (b) Fertilization
(c) Ovulation (d) Implantation

Answer: (c) Ovulation — It happens once each month.

24. After ovulation, the ruptured follicle changes into the:

(a) Graafian follicle (b) Polar body
(c) Placenta (d) Corpus luteum

Answer: (d) Corpus luteum — It is a glandular structure.

25. Oviducts are also called:

(a) Fallopian tubes (b) Ureters
(c) Vasa deferentia (d) Ejaculatory ducts

Answer: (a) Fallopian tubes — They are also called uterine tubes.

26. The ovum moves through the oviduct by the action of:

(a) Flagella (b) Cilia
(c) Muscles of ovary (d) Gravity

Answer: (b) Cilia — Cilia line the oviduct.

27. The uppermost rounded portion of the uterus is the:

(a) Cervix (b) Vagina
(c) Body (d) Fundus

Answer: (d) Fundus — It lies above the openings of the oviducts.

28. The lower narrow part of the uterus is the:

(a) Cervix (b) Body
(c) Fundus (d) Endometrium

Answer: (a) Cervix — Its opening leads to the vagina.

29. The middle thick muscular layer of the uterus wall is the:

(a) Perimetrium (b) Endometrium
(c) Myometrium (d) Epithelium

Answer: (c) Myometrium — It is made of smooth muscles.

30. The layer of the uterus shed during menstruation is the:

(a) Basal layer of endometrium (b) Perimetrium
(c) Myometrium (d) Functional layer of endometrium

Answer: (d) Functional layer of endometrium — The basal layer regenerates it.

31. The structure that supplies oxygen and nutrition to the foetus is the:

(a) Corpus luteum (b) Placenta
(c) Cervix (d) Fundus

Answer: (b) Placenta — It forms from the endometrium.

32. The length of the vagina is about:

(a) 8–10 cm (b) 3–5 cm
(c) 15–20 cm (d) 25 cm

Answer: (a) 8–10 cm — It is a thin-walled tube.

33. Oogonia divide by mitosis to produce:

(a) Secondary oocytes (b) Polar bodies
(c) Primary oocytes (d) Ova

Answer: (c) Primary oocytes — This occurs during embryonic development.

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34. Primary oocytes stop division in:

(a) Prophase-I (b) Metaphase-I
(c) Metaphase-II (d) Anaphase-II

Answer: (a) Prophase-I — They remain so all through childhood.

35. The secondary oocyte stops division in:

(a) Prophase-I (b) Metaphase-II
(c) Anaphase-I (d) Telophase-II

Answer: (b) Metaphase-II — It completes meiosis-II only after sperm entry.

36. Meiosis-I of a primary oocyte produces a secondary oocyte and a:

(a) Spermatid (b) Second polar body
(c) First polar body (d) Follicle

Answer: (c) First polar body — The polar body is the smaller cell.

37. At birth, a female's ovaries contain about:

(a) 2 thousand follicles (b) 20 million follicles
(c) 200 follicles (d) 2 million follicles

Answer: (d) 2 million follicles — Each has a primary oocyte.

17.1.3 THE MENSTRUAL CYCLE

38. The average length of the menstrual cycle is:

(a) 14 days (b) 21 days
(c) 28 days (d) 40 days

Answer: (c) 28 days — It lasts approximately one month.

39. Menstruation occurs due to low levels of:

(a) FSH and LH (b) Estrogen and progesterone
(c) GnRH (d) hCG

Answer: (b) Estrogen and progesterone — The functional layer of endometrium is shed.

40. The menstrual phase covers days:

(a) 1–5 (b) 6–14
(c) 15–28 (d) 10–20

Answer: (a) 1–5 — Bleeding continues for 3–5 days.

41. The mature ovarian follicle is called the:

(a) Corpus luteum (b) Placenta
(c) Polar body (d) Graafian follicle

Answer: (d) Graafian follicle — Only one follicle grows fully.

42. FSH stimulates the Graafian follicle to secrete:

(a) Progesterone (b) LH
(c) Estrogen (d) hCG

Answer: (c) Estrogen — Estrogen makes the endometrium thick again.

43. Ovulation occurs on about day:

(a) 5 (b) 14
(c) 21 (d) 28

Answer: (b) 14 — It follows the sudden release of LH.

44. The corpus luteum secretes:

(a) FSH and LH (b) Inhibin
(c) hCG (d) Estrogen and progesterone

Answer: (d) Estrogen and progesterone — They make the endometrium vascular and glandular.

45. If fertilization does not occur, the corpus luteum:

(a) Degenerates (b) Grows larger
(c) Becomes the placenta (d) Releases an ovum

Answer: (a) Degenerates — Estrogen and progesterone then fall.

46. The hormone tested in pregnancy tests is:

(a) LH (b) FSH
(c) Estrogen (d) hCG

Answer: (d) hCG — It comes from the embryonic chorion.

47. hCG maintains the:

(a) Corpus luteum (b) Graafian follicle
(c) Primary oocyte (d) Cervix

Answer: (a) Corpus luteum — So menstruation is prevented.

48. The stopping of the menstrual cycle around 50 years of age is called:

(a) Puberty (b) Ovulation
(c) Menopause (d) Gestation

Answer: (c) Menopause — It marks the end of the reproductive period.

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